

emem: A research on Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

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If Agent A cites an Earth observation, can Agent B recover **exactly what was cited** after the context is gone?

0.4871541501976284 became “≈ 0.49”. The decision flipped.

REFERENTIAL DRIFT. WATER if NDVI < 0.488. Exact token: both models WATER. Paraphrase: both models SKIP.

Keep the scene upstream. Move the address into reasoning. A later agent resolves the cited record, re-hashes the bytes, checks the attestation, then reasons from the same evidence.

LOCATE → MATERIALISE → ADDRESS → RESOLVE → VERIFY → GUARD → HAND OFF
place → cell64 → observation/field → token → exact bytes → verdict → token/bundle

TWO KEYS, TWO JOBS

LOOK UP

cell64 × band × tslot
where - what variable - when

IDENTIFY

fact_cid = BLAKE3(emem-CBOR(record))
exactly which signed observation record

HAND OFF

emem:fact:<cell64>:<fact_cid>
location binding + immutable record identity
Spatial/temporal indexing finds the observation. Content addressing keeps its identity stable after it enters model reasoning.

Fig. 1 — One real observation: source → record → content id → handoff

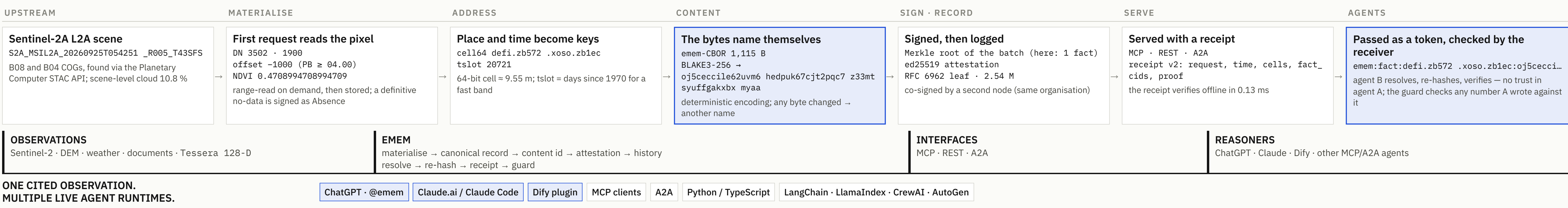


FIG. 2A — TRUE COLOUR, FROM SIGNED GRIDS



B04/B03/B02, each an emem:raster: artifact of 802,780 B re-hashed ✓ · 443 × 453 px at 10 m · 25 Sep 2026

FIG. 2B — NDVI COMPUTED BY US FROM B08, B04

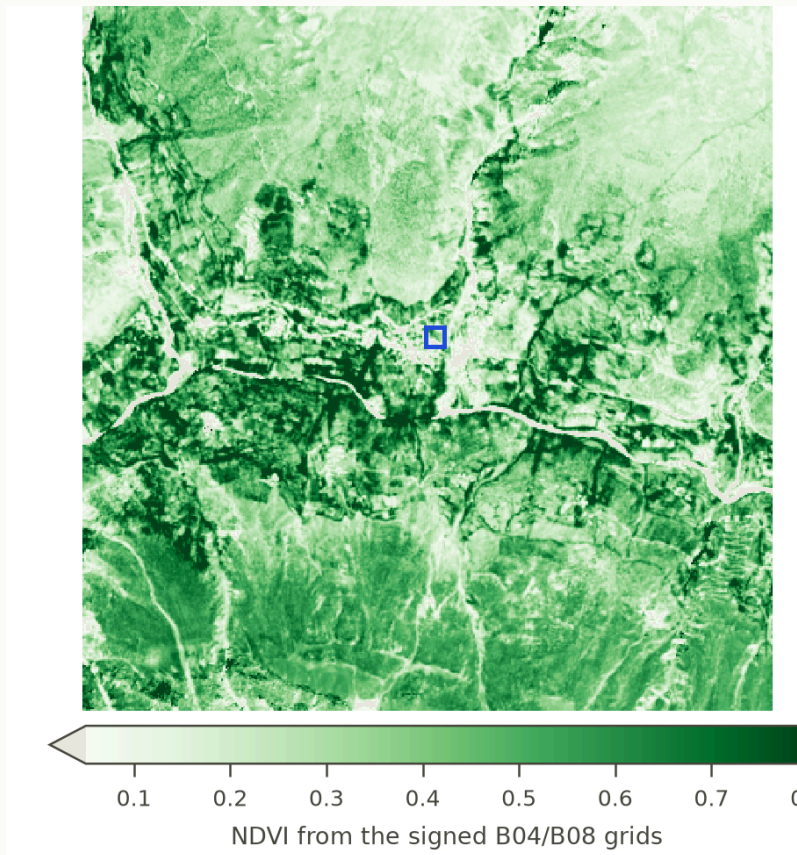
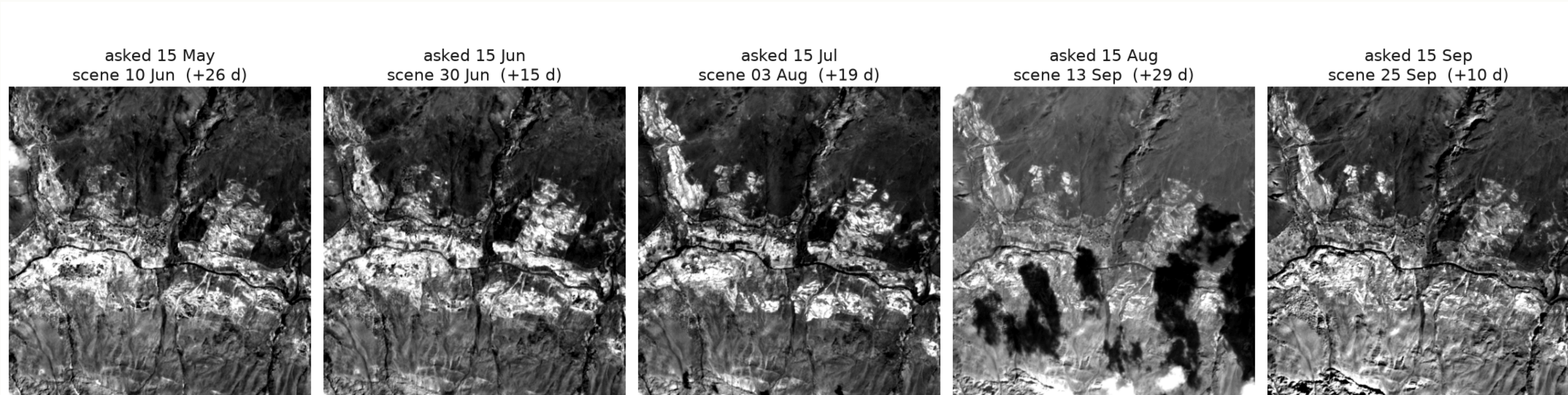


FIG. 2C — FIELD — TIME: ONE CUBE TOKEN, FIVE INDEPENDENTLY RESOLVABLE SCENES



One emem:cube: token, five members, each its own raster token and re-hashed artifact. Asked for the 15th of each month; the scene actually used is 10–29 days later, and the manifest says so. The 13 Sep member is cloudy: selection uses scene-level cloud cover (≤ 40 %, relaxed in tiers), not a per-pixel mask — a limit we show rather than crop.

01 CONTENT ID

The observation record is (cell, band, tslot, value, uncertainty, sources, derivation, signer, signed_at). Deterministic emem-CBOR is hashed with BLAKE3; the served bytes must re-hash to the id.

```
REAL KEYLONG NDVI · VERIFY_NDVI.PY
fact 0j5cecci... 1,115 bytes re-hash MATCH
B08 DN 3502 B04 DN 1900 offset -1000
signed 0.4708994708994709
rebuilt 0.4708994708994709 bit-identical
same cid under another cell → HTTP 409
```

CHECK changed bytes → changed id; wrong place → HTTP 409. The same digest is also emitted as CIDv1 for generic content-address tooling.

BOUNDARY the id names one signed record because signer and signed_at are inside it; it is not a universal “truth id”.

03 FIELD / CUBE / EMBEDDING

emem:raster:: names a spatial field; emem:cube: binds fields through time. The live Tesserà encoder’s 128-D output is stored as a typed model_output fact: the embedding is a value inside an addressable record, not the memory itself.

```
REAL KEYLONG B08 CUBE · FIG. 2
15 May → 10 Jun (+26 d)
15 Jun → 30 Jun (+15 d)
15 Jul → 03 Aug (+19 d)
15 Aug → 13 Sep (+29 d)
15 Sep → 25 Sep (+10 d)
5 artifacts re-hashed ✓
```

FLOW Sentinel-2 → field / Tesserà 128-D → signed record → token → any connected agent runtime.

BOUNDARY scene selection uses scene-level cloud cover; re-minting can produce a new derivation id for identical pixels.

02 HANDOFF

Agent B does not need Agent A’s transcript to recover the evidence. It needs the token. For many cited observations, **emem:bundle:** carries up to 256 fact references under one compact handle.

```
MEASURED PARAPHRASE TRAP · ONE REAL LAHAUL FACT
signed NDVI 0.4871541501976284
decision rule WATER if NDVI < 0.488
token handoff Gemma WATER · Qwen WATER correct
prose “≈ 0.49” Gemma SKIP · Qwen SKIP agree + wrong
```

LONG RUN bundle: ≤256 facts · 38 characters · ~23 LLM tokens. In the measured 599-turn run, bundle answers were 100% citable vs BM25 0%.

BOUNDARY one fact token is ~51 LLM tokens; bundling—not a single token—is the context-saving mechanism.

04 CLAIM CHECK

guard/verdict resolves citations in a draft and compares the written number with the signed value at the writer’s precision. A mismatch returns a typed reason and fix.

```
REAL BENGALURU TEMPERATURE FACT = 28.0 °C
“... was 35 °C (token)” deny PROV_VALUE
“... was 28.0 °C (token)” allow
same cid under another cell deny PROV_BYTES
uncited measurable claim, gating deny CLAIM_UNGROUNDED
```

LOOP OBSERVE → ADDRESS → HAND OFF → RESOLVE → VERIFY → CONTINUE.

BOUNDARY the hosted guard is advisory; it does not yet check whether the cited band semantically matches the sentence.

STAC / COG

asset / raster file
Discover and read EO assets.

openEO / Earth Engine

process / computed object
Execute geospatial analysis.

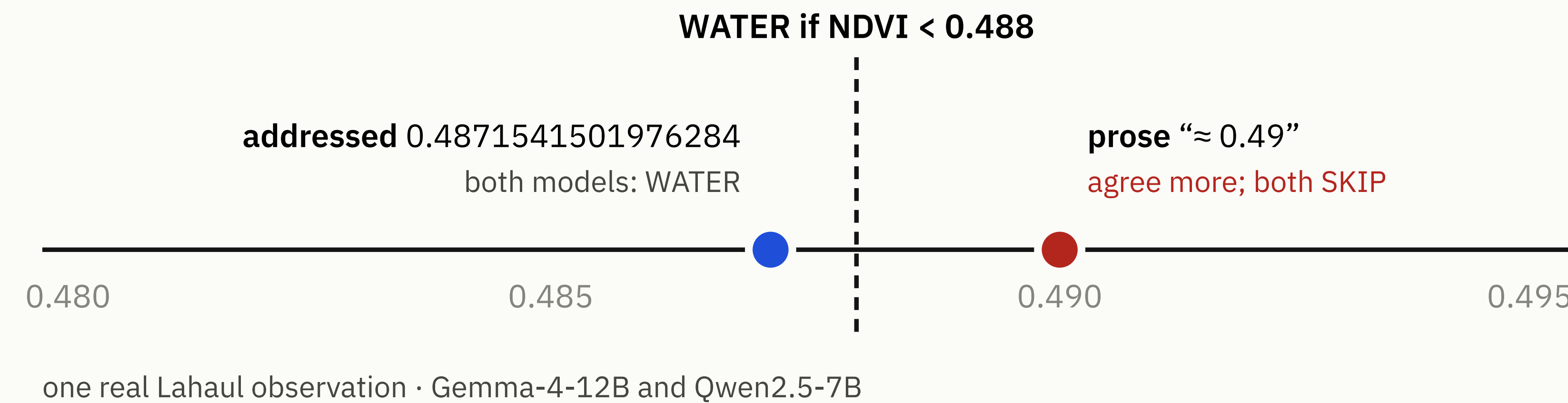
RAG / agent memory

retrieved context
Recover useful text or task state.

emem

observation / field address
Resolve the cited physical-world record and check its bytes/attestation.

DRIFT · agreement can rise while correctness falls



full-context control 72/72 correct · 36/36 agree

shared compaction summary 0/72 correct · 3/36 agree · p = 0.035

20-turn handoff: prose · dense · BM25 · bundle 2 · 8 · 20 · 20 correct

599-turn run: citable, bundle vs BM25 100% vs 0%

Pre-registered result; Fisher one-sided. Two open models on one host; no independent replication. The known phenomenon is lossy multi-agent communication. emem’s contribution is a handoff object whose cited evidence can be recovered and checked instead of paraphrased.

TEST THE HYPOTHESIS LIVE

```
# one checkable object
TOKEN='emem:fact:...'

# resolve exact bytes
curl .../memory_token/resolve

# recompute its BLAKE3 id
python verify_fact.py "$TOKEN"

# verify the signed receipt offline
python verify_receipt_tamper.py receipt_2026-09-29.json

# then hand the same token to another agent
```

WHY HISTORY MATTERS. The same Bengaluru elevation key served **918.0 m** in May and **915.07 m** after the upstream changed. Both old and new records still re-hash to their own CIDs, so a later agent can recover what an earlier agent actually cited.

What holds: byte identity, spatial binding, signer/receipt integrity, replayable cited state.

What does not: objective truth, sensor accuracy, model correctness or a guarantee that the prose conclusion follows from the cited band.

The poster uses real tokens, real Sentinel-2 grids and independently re-run checks. Detailed prior art, audit defects, withdrawn claims and all reproduction scripts stay in the repository rather than competing for wall space.